

Critical Slowing Down

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Abstract

This project explores critical slowing down in the two-dimensional Ising model near the critical temperature T_c . Using Markov chain Monte Carlo methods, we implement both the Metropolis and Wolff algorithms to generate spin configurations and measure autocorrelation times of magnetization and energy. Magnetic susceptibility is determined with an external field B , and, in the absence of the field, via the fluctuation–dissipation theorem (FDT). The scaling of relaxation times allows estimation of the dynamic critical exponent z for each algorithm. Comparisons between local and cluster updates highlight their relative efficiency at criticality, showing how algorithmic choices affect the sampling of Markov chains in systems exhibiting critical slowing down and field-induced responses.

1 Introduction: Dynamic Critical Exponents and Autocorrelation Times

Understanding dynamic properties of Monte Carlo simulations requires treating the update dynamics as a stochastic process. In this framework, the time evolution of a Monte Carlo configuration is described by a Markov chain. This provides the natural foundation for defining autocorrelation functions, relaxation times, and ultimately the dynamic critical exponent z .

1.1 Markov Chains and Master Equations

For relaxational models, such as the stochastic Ising model, the time-dependent behavior is described by a master equation

$$\frac{\partial P_n(t)}{\partial t} = - \sum_{n \neq m} [P_n(t)W_{n \rightarrow m} - P_m(t)W_{m \rightarrow n}] \quad (1)$$

where $P_n(t)$ is the probability of the system being in state n at time t , and $W_{n \rightarrow m}$ is the transition rate for $n \rightarrow m$. The solution to the master equation is a sequence of states, but the time variable is a stochastic quantity which does not represent true time. A relaxation function $\phi(t)$ can be defined which describes time correlations *within equilibrium*

$$\phi_{MM}(t) = \frac{\langle M(0)M(t) \rangle - \langle M \rangle^2}{\langle M^2 \rangle - \langle M \rangle^2}. \quad (2)$$

When normalized in this way, the relaxation function is 1 at $t = 0$ and decays to zero as $t \rightarrow \infty$. It is important to remember that for a system in equilibrium any time in the sequence of states may be chosen as the ‘ $t = 0$ ’ state. The asymptotic, long time behavior of the relaxation function is exponential, i.e.

$$\phi(t) \rightarrow e^{-t/\tau}. \quad (3)$$

where the correlation time τ diverges as T_c is approached. This dynamic (relaxational) critical behavior can be expressed in terms of a power law as well,

$$\tau \propto \xi^z \propto \varepsilon^{-\nu z}. \quad (4)$$

where ξ is the (divergent) correlation length, $\varepsilon = |1 - T/T_c|$, and z is the dynamic critical exponent.

In a finite system of linear size L , the divergence of the correlation length at criticality is limited by the system size. Exactly at T_c , the correlation length therefore saturates at $\xi \sim L$. Inserting this finite-size cutoff into the critical slowing-down relation $\tau \propto \xi^z$ gives

$$\tau_c \sim L^z. \quad (5)$$

Thus, at the critical temperature, the relaxation time is no longer controlled by the reduced temperature ε , but instead by the system size itself. As L increases, correlations require more time to propagate across the entire system, leading to a relaxation time that scales as a power law in L with exponent z .

A second relaxation time, the integrated relaxation time, is defined by the integral of the relaxation function

$$\tau_{\text{int}} = \int_0^\infty \phi(t) dt. \quad (6)$$

This quantity has particular importance for the determination of errors and is expected to diverge with the same dynamic exponent as the ‘exponential’ relaxation time.

2 Metropolis Algorithm for the 2D Ising model

Metropolis method (**ref**) configurations are generated from a previous state using a transition probability which depends on the energy difference between the initial and final states. The states produced follows a time ordered path, and this time is not a proper time but a Monte-Carlo

time. For relaxational models, such as the stochastic Ising model, the time-dependent behavior is described by the master equation (1) :

$$\frac{\partial P_n(t)}{\partial t} = - \sum_{n \neq m} [P_n(t)W_{n \rightarrow m} - P_m(t)W_{m \rightarrow n}] \quad (7)$$

with the probability of the n th state occurring in a classical system is given by

$$P_n = \frac{e^{-\beta E_n}}{Z}$$

with Z the partition function which we don't know about. To bypass the expression of Z , we generate a Markov chain of states, i.e. we generate each new state directly from the preceding state. If we produce the n_{th} state from the m_{th} state, and then calculate the relative probability which is the ratio of the individual probabilities, the denominator cancels. As a result, only the energy difference between the two states is needed : $\Delta E = E_n - E_m$

Indeed know thanks to the master equation (1) that in equilibrium :

$$P_n W_{n \rightarrow m} = P_m W_{m \rightarrow n}$$

Thus

$$\frac{W_{n \rightarrow m}}{W_{m \rightarrow n}} = \frac{P_m}{P_n} = e^{-\beta \Delta E}$$

Therefore,

$$W_{n \rightarrow m} = \begin{cases} \tau_0^{-1} e^{-\beta \Delta E} = e^{-\beta \Delta E} & \text{si } \Delta E > 0, \\ \tau_0^{-1} = 1 & \text{si } \Delta E \leq 0. \end{cases}$$

as we know that $W_{n \rightarrow m}$ is of the dimensional of 1/time and τ_0 is the time required to attempt a spin-flip that we choose equal to 1.

The steps of the algorithm are as follows:

1. Choose an initial state.
2. Choose a lattice site i .
3. Compute the energy change ΔE resulting from flipping the spin at site i .
4. Generate a random number r such that $0 < r < 1$.
5. If $r < e^{-\beta \Delta E}$, flip the spin.
6. Move to the next site and return to step (3).

2.1 Single-Spin Metropolis algorithm (checkerboard) on 2D Ising Model

In this work we use a *hot start* (random initial condition) for each run: the lattice spins are initialized independently to +1 or -1 with equal probability. The hot start avoids biasing the system toward a magnetized state and is particularly convenient when scanning temperatures around T_c .

We employ a single-spin *Metropolis* algorithm with a checkerboard (updating parity) scheme. The checkerboard scheme divides the square lattice into two sublattices (even/odd parity) so that all spins updated in the same micro-step have only neighbours belonging to the opposite parity; this allows an explicit, conflict-free update sequence:

for parity = 0, 1 : update all sites (i, j) with $(i + j) \bmod 2 = \text{parity}$.

Periodic boundary conditions are used throughout. For a proposed flip of spin $s_{i,j}$ the energy change is

$$\Delta E = 2s_{i,j} \left(J \sum_{\langle k \rangle} s_k + h \right),$$

and the Metropolis acceptance probability is

$$P_{\text{accept}} = \min(1, e^{-\beta \Delta E}).$$

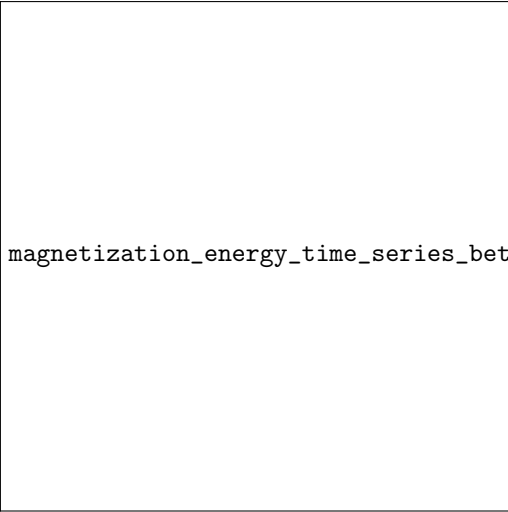
2.2 Monte Carlo time, equilibration and measurements

The algorithm produces a time-ordered sequence of configurations; the index along this sequence is the *Monte Carlo time* (in units of sweeps when we define one sweep = $L \times L$ single-spin update attempts). Because this time is algorithmic rather than physical, we refer to relaxation times in units of Monte Carlo sweeps.

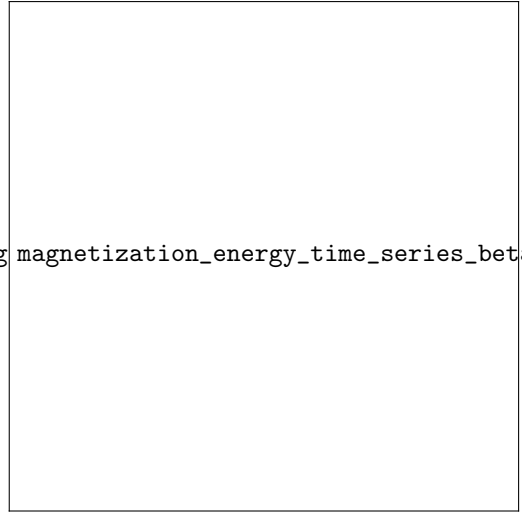
Before recording measurements we perform an equilibration phase of several thousand sweeps (or more near T_c) to ensure the system has reached stationarity. After equilibration we record observables (energy, total magnetization) at regular intervals. The choice of recording interval should be made in view of the autocorrelation time (below) to reduce correlation between successive measurements.

Plotting the energy and magnetization time series for various inverse temperatures β allows us to verify that the system reaches equilibrium, identify enhanced fluctuations near the critical temperature β_c , and observe critical slowing down. These plots are also used to determine the appropriate *burn-in* period, i.e., the number of initial sweeps to discard before measurements, and to choose suitable recording intervals between successive measurements, ensuring that the collected data are sufficiently decorrelated.

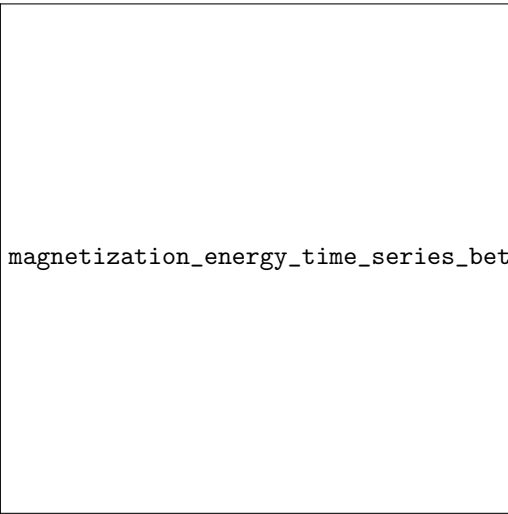
At low values of β (high temperatures), both energy and magnetization fluctuate around mean values with relatively small amplitudes, indicating that the system is in a disordered phase and has reached equilibrium quickly. As β approaches the critical region near $\beta_c \approx 0.44$, fluctuations in magnetization become much larger, reflecting the increased susceptibility and the onset of critical behavior. The energy time series remains relatively stable, but subtle increases in fluctuations can also be observed. Beyond β_c , magnetization tends to spend more time near nonzero values, corresponding to the emergence of spontaneous magnetization in the ordered phase. These plots also illustrate critical slowing down, as configurations near β_c require longer Monte Carlo times to decorrelate, highlighting the need for a sufficiently long burn-in and appropriately spaced measurements.



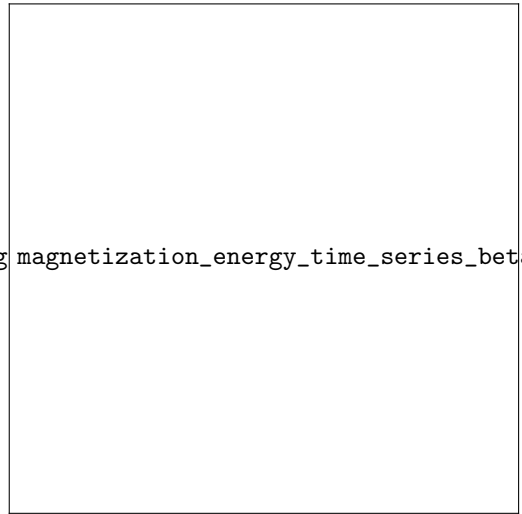
(a) $\beta = 0.1$



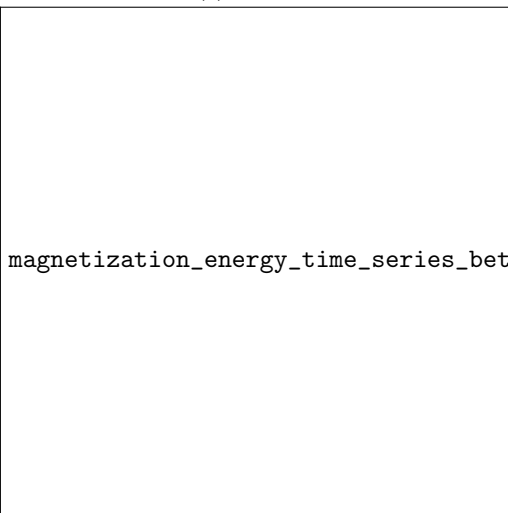
(b) $\beta = 0.2$



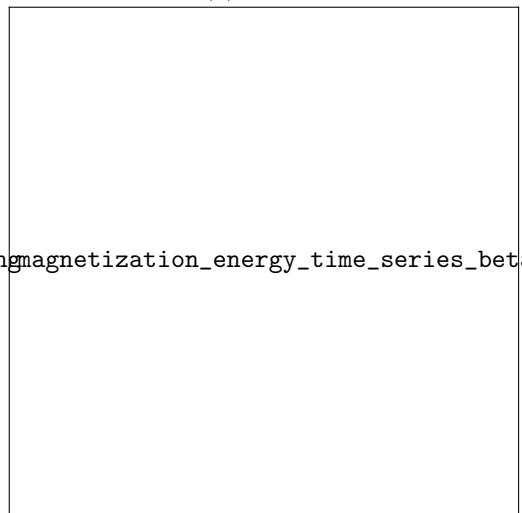
(c) $\beta = 0.3$



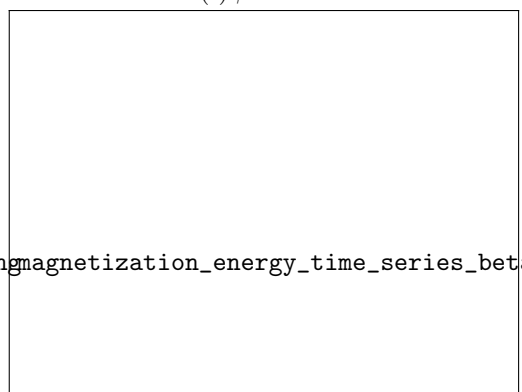
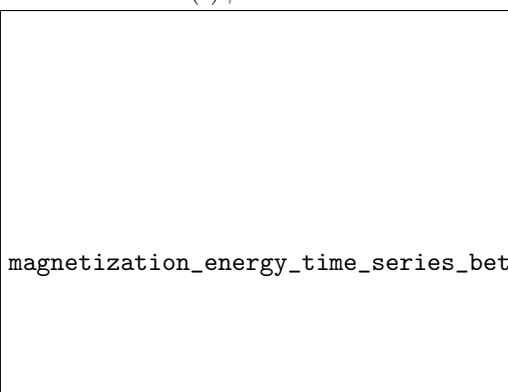
(d) $\beta = 0.4$

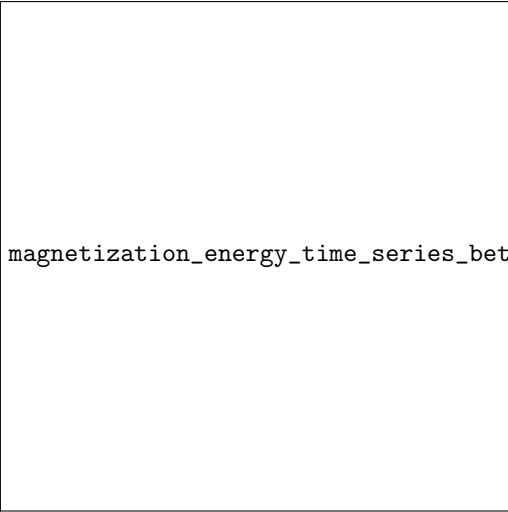


(e) $\beta = 0.41$



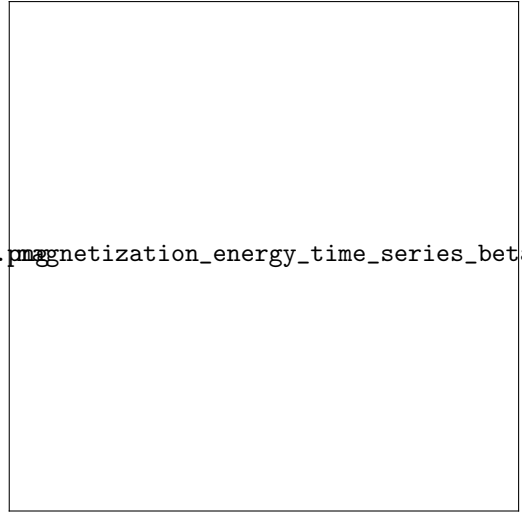
(f) $\beta = 0.42$





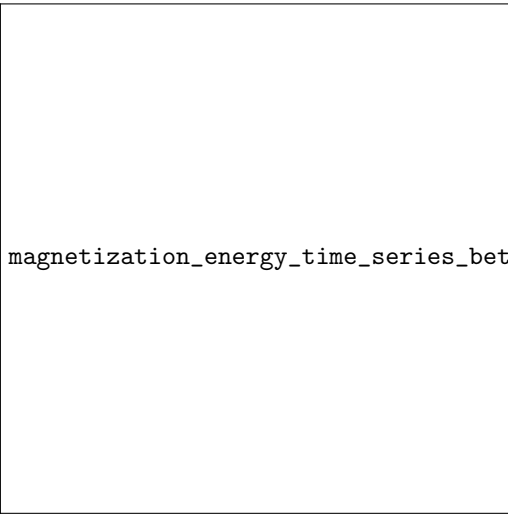
magnetization_energy_time_series_beta0.4407.png

(a) $\beta = 0.4407$



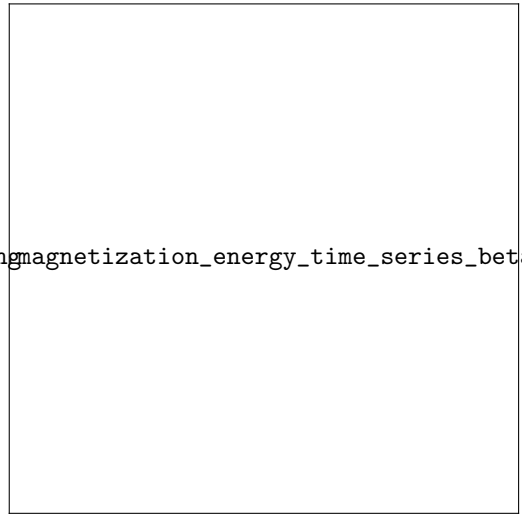
magnetization_energy_time_series_beta0.45.png

(b) $\beta = 0.45$



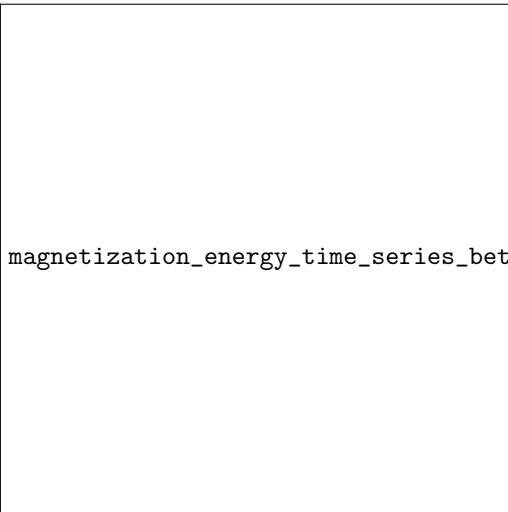
magnetization_energy_time_series_beta0.46.png

(c) $\beta = 0.46$



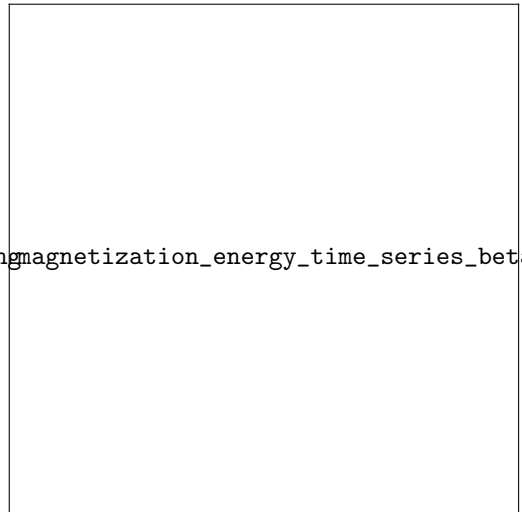
magnetization_energy_time_series_beta0.47.png

(d) $\beta = 0.47$



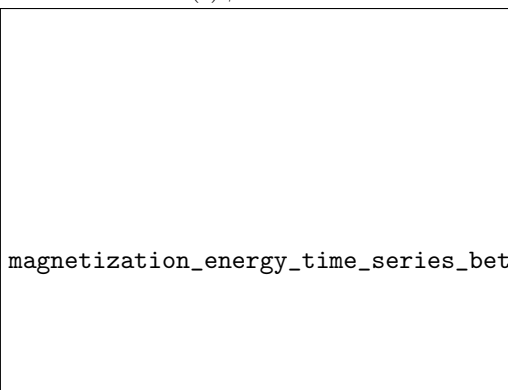
magnetization_energy_time_series_beta0.48.png

(e) $\beta = 0.48$

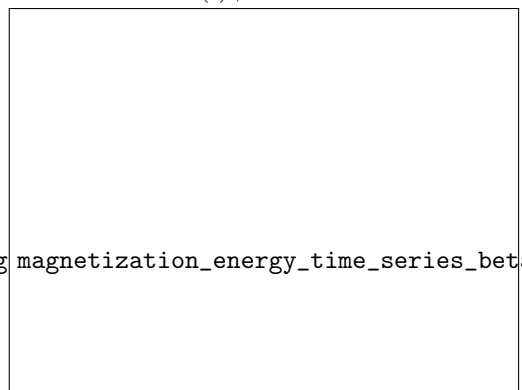


magnetization_energy_time_series_beta0.49.png

(f) $\beta = 0.49$



magnetization_energy_time_series_beta0.5.png



magnetization_energy_time_series_beta0.6.png

2.3 Autocorrelation and integrated autocorrelation time

To quantify critical slowing down we compute the autocorrelation function of an observable A (for instance the magnetization M , the energy or the susceptibility):

$$C_A(t) = \langle A_s A_{s+t} \rangle - \langle A \rangle^2,$$

and the normalized autocorrelation (autocorrelation coefficient)

$$\rho_A(t) = \frac{C_A(t)}{C_A(0)}.$$

A convenient scalar measure of the memory of the Markov chain is the *integrated autocorrelation time*

$$\tau_{\text{int},A} = \frac{1}{2} + \sum_{t=1}^{T_{\text{max}}} \rho_A(t),$$

where the sum is truncated at a cutoff T_{max} chosen e.g. by the window method (or until $\rho_A(t)$ has decayed to noise). The statistical error of a sample mean $\bar{A}$ computed from N recorded measurements is increased by autocorrelation; the effective sample size is $N_{\text{eff}} \approx N/(2\tau_{\text{int},A})$.

2.4 Extracting the dynamic exponent z

At criticality finite-size scaling for relaxational dynamics predicts that the characteristic relaxation time of the system at T_c scales with the linear size L as

$$\tau_c(L) \propto L^z,$$

because the correlation length is cut off by the system size $\xi \sim L$ and $\tau \propto \xi^z$. In practice we estimate $\tau_c(L)$ by the integrated autocorrelation time of the magnetization (or by the exponential autocorrelation time obtained from a fit of $\rho(t) \approx e^{-t/\tau_{\text{exp}}}$), and then fit $\log \tau_c$ versus $\log L$ to extract z :

$$\log \tau_c = z \log L + \text{const.}$$

2.5 Susceptibility and measurement formulas

For a system of $N = L^2$ spins and inverse temperature β the magnetic susceptibility at zero field is estimated from recorded magnetizations M_k as

$$\chi = \beta N (\langle m^2 \rangle - \langle |m| \rangle^2), \quad m = \frac{M}{N}.$$

When a small nonzero external field h is present one should instead use $\langle m \rangle$ (not $\langle |m| \rangle$) in the above variance term.

2.6 Error estimation

To estimate statistical uncertainties we recommend the blocking (binning) method: group consecutive measurements into blocks of size B (with $B \gg \tau_{\text{int}}$), compute the block means and use the sample variance of block means to estimate the variance of the mean. The block size should be increased until the estimated error stabilizes.

3 Wolff Algorithm for the 2D Ising model

Wolff (**ref**) proposed an alternative algorithm based on the Fortuin–Kasteleyn theorem in which single clusters are grown and flipped sequentially. The algorithm begins with the random choice of a single site. Bonds are then drawn to all nearest neighbors which are in the same state with probability

$$p = 1 - e^{-K\delta_{\sigma_i}\sigma_j}$$

One then moves to all sites in turn which have been connected to the initial site and places bonds between them and any of their nearest neighbors which are in the same state with probability given by equation (6). The process continues until no new bonds are formed and the entire cluster of connected sites is then flipped. Another initial site is chosen and the process is then repeated.

- 3.1 Measure of autocorrelation functions of magnetization, energy and susceptibility
- 3.2 Measure of autocorrelation times near T_c
- 3.3 Estimation of the dynamic critical exponent z
- 4 Comparison of algorithm efficiency at criticality